Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

ACTIVITY 1 — TRAINER PACK

For

ISO 14001:2015 Internal Auditor Training

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Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

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Activity 1 — Auditing GreenPack — Framework, Context and Scope

Topic 1 · **45 minutes** · Assesses **K1 · K2 · A1** · Course TGS-2025054479

Learning purpose

GreenPack Manufacturing has expanded capacity by 40% and added a solvent-based thermal lamination line. Energy use is up 25%, VOC emissions have risen and three odour complaints have been received — but the aspects register and policy are stale. You are the lead internal auditor: scope the engagement.

Scenario

GreenPack Manufacturing Pte Ltd is a Singapore company producing biodegradable packaging materials from a single leased facility in Tuas. It achieved ISO 14001:2015 certification two years ago and employs 150 staff across three departments: Production, Warehousing & Logistics, and Administration.

Eight months ago the company expanded production capacity by 40% and introduced a new **thermal lamination process using solvent-based adhesives**. Since the expansion:

- electricity consumption has risen by 25%;
- volatile organic compound (VOC) emissions from the lamination line have increased;
- **three customer complaints** about chemical odours from the packaging have been received;
- one **neighbouring tenant** has complained informally to the landlord about solvent smell.

The Environmental Policy was last reviewed 18 months ago and makes no reference to the new process. The environmental aspects and impacts register has **not been updated since the expansion**. The company's strategic plan carries an objective to **reduce its carbon footprint by 30% within three years**. The internal audit function follows the **International Professional Practices Framework (IPPF)**.

You are the lead internal auditor. The Managing Director has directed the internal audit team to conduct a comprehensive EMS audit and wants the scope agreed this week. You have been given the Environmental Policy, the (stale) aspects register, the organisation chart and the strategic plan.

The evidence you have been given

- Certified to ISO 14001:2015 for 2 years; single leased site in Tuas; 150 staff.
- Capacity +40%; new solvent-based thermal lamination line commissioned 8 months ago.
- Electricity consumption +25%; VOC emissions increased.
- 3 customer odour complaints; 1 informal neighbour complaint via the landlord.
- Environmental Policy last reviewed 18 months ago — does not mention lamination.
- Aspects & impacts register not updated since the expansion.
- Strategic objective: reduce carbon footprint 30% in 3 years.
- Internal audit function operates under the IPPF.
- You report to the Audit Manager; you have never worked in Production.

What you will produce

DELIVERABLE — A completed Audit Engagement Scoping Note for GreenPack — clause map,

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context table, prioritised scope and the IPPF basis for your independence.

Step-by-step method

Step 1: Establish your professional basis under the IPPF

List the four mandatory IPPF components and state, in one line each, how they govern THIS engagement: the Definition of Internal Auditing (independent, objective assurance adding value); the Code of Ethics (integrity, objectivity, confidentiality, competency); the Core Principles; and the International Standards. Note explicitly that you have never worked in Production — record that your independence is intact.

Step 2: Convert each symptom into a clause hypothesis

Take every fact in the scenario and ask 'which ISO 14001:2015 requirement would have prevented this?'. Stale policy -> 5.2. Stale aspects register after a process change -> 6.1.2 (change is explicitly named). Odour and VOC -> 6.1.2, 8.1 operational control, 6.1.3 compliance obligations. Neighbour and customer complaints -> 4.2 interested parties and 7.4 communication. Energy +25% against a carbon objective -> 6.2, 9.1.1.

Step 3: Build the context table (clause 4.1 / 4.2)

Three columns: EXTERNAL issues (NEA air-quality regulation, landlord lease terms, customer sustainability expectations, solvent price), INTERNAL issues (new process, capacity change, competence of new operators, ageing policy), and INTERESTED PARTIES with their relevant needs (NEA, landlord, neighbouring tenants, customers, employees). Then apply clause 4.2(c): mark which of those needs the organisation has ADOPTED as compliance obligations.

Step 4: Test the declared EMS scope (clause 4.3)

Ask whether the certified scope still covers what the company actually does. A 40% capacity increase and a brand-new process either fall inside the existing scope statement or reveal that the scope was never updated. Record which, and the evidence you would request to confirm it.

Step 5: Prioritise the audit by risk, not by department

Rank the areas: lamination line and its emissions (highest — new, uncontrolled, complaints, legal exposure); aspects/risk management process; policy and management review; energy and the carbon objective; warehousing; administration (lowest). State the reason for each rank in one line — this is what makes the plan defensible.

Step 6: Write the engagement scoping note

One page: audit objective, criteria (ISO 14001:2015 + GreenPack's own EMS procedures + applicable compliance obligations), scope and boundaries, areas prioritised with rationale, auditor independence statement, and the documents you will request before fieldwork.

Questions to answer

Q1. Which IPPF components govern your conduct on this engagement, and what does each require of you here?

What a good answer contains: Expect all four mandatory elements named, each tied to a concrete action in this case (e.g. Objectivity — you must not audit work you performed; Competency — you may need technical support on solvent chemistry).

Q2. Which ISO 14001:2015 clauses are most relevant to GreenPack's situation, and why is each applicable?

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What a good answer contains: Look for 6.1.2 (change / stale register) and 5.2 (stale policy) as the two strongest, plus 4.1, 4.2, 4.3, 6.1.3, 6.2, 7.4, 8.1, 9.1.1 and 9.3, each justified by a specific fact from the scenario — not a generic clause list.

Q3. How should GreenPack's risks, objectives, resources and operational changes shape the scope and focus of your audit plan?

What a good answer contains: The answer must move from risk to scope: the highest-risk, most-changed, most-complained -about area gets the most audit time. Equal time per department is the wrong answer.

Self-check

YOU ARE DONE WHEN — Your scoping note names at least 8 relevant clauses each tied to a specific scenario fact, ranks the lamination line as the top priority with a stated reason, and contains an explicit auditor-independence statement.

If you are stuck — what usually goes wrong

- You listed clauses but did not tie each one to a fact from the scenario — the clause map must be evidence-led, not a contents page.
- You allocated equal audit time to all three departments. Audit effort follows environmental risk and change; say so and defend it.
- You treated the odour complaints as the finding. They are a SYMPTOM — the stale aspects register is the systemic issue.
- You forgot the independence statement. The IPPF requires objectivity to be considered and recorded before the engagement starts.
- You wrote 'the policy is old' without naming clause 5.2 and what it requires.

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Model answer (trainer only)

IPPF. The Definition frames the work as independent, objective assurance that adds value. The **Code of Ethics** binds the auditor to Integrity, Objectivity, Confidentiality and Competency — here, objectivity is satisfied because the auditor has never worked in Production, and competency may require a technical specialist on solvent emissions. The **Core Principles** and the **International Standards** require the engagement to be risk-based, adequately planned and reported to those charged with governance.

Clause map. 4.1 — the new process and rising energy cost are internal issues; NEA air regulation and the landlord's lease are external. 4.2 — neighbours, customers, NEA and the landlord are interested parties whose expectations may become compliance obligations under 4.2(c). 4.3 — the EMS scope must still be valid after a 40% expansion. **5.2 — the policy has not been reviewed since a material process change and does not cover lamination: a likely nonconformity. 6.1.2 — the standard explicitly requires aspects to be determined considering change, including new or modified activities; the register is 8 months out of date: a likely major.** 6.1.3 — VOC discharge triggers EPMA/air-impurities obligations that must be identified. 6.2 — the 30% carbon objective is contradicted by a 25% energy rise. 7.4 — three customer complaints and a neighbour complaint test the communication process. 8.1 — operational control and operating criteria for the new line. 9.1.1 — is VOC and energy performance being monitored at all? 9.3 — did management review consider the change?

Scope. Objective: determine whether the EMS conforms to ISO 14001:2015 and to GreenPack's own requirements, and is effectively implemented, with particular regard to the capacity expansion and the lamination process. Boundaries: the Tuas site, all three departments, EMS activity from the date of the expansion. Priority order: (1) lamination line — aspects, controls, emissions, legal; (2) the aspect-identification and change-management process itself; (3) policy, objectives and management review; (4) energy performance vs the carbon objective; (5) warehousing; (6) administration. Equal time per department would be indefensible: audit effort follows environmental risk and change.

Trainer debrief

Draw out three points. First, **the stale aspects register is the systemic finding** — everything else (odour, VOC, complaints) is a symptom of a change-management process that does not trigger a re-evaluation of aspects. Second, **clause 6.1.2 names 'change' explicitly**, so this is not an auditor's opinion but a requirement breach. Third, **scope is a risk decision**: the group that allocated equal time to all three departments should defend that against the group that front-loaded the lamination line.

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